25901398 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCMBG201 - FOOD MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

PART A

Short Notes Questions

Answer any **10** questions

Each question carries **2** marks

1. **List the factors influencing kind and rate of spoilage in fish and other seafoods** **[K] / [CO1]**
   Ans: Any 2 factors.(2 marks)
2. **Name one bacterium that causes sourness in beer.** **[K] / [CO2]**
   Ans: Species of Acetobacter or Glucanobacter (2 marks)
3. **Mention two symptoms of Trichinosis.** **[K] / [CO3]**
   Ans: Any two symptoms. (2 marks)
4. **Write notes on anthracnose.** **[K] / [CO1]**
   Ans: Definition (2 marks)
5. **What brine condition favors black pickles?** **[K] / [CO2]**
   Ans: Low acidity of the brines.(2 marks)
6. **Which carbohydrate undergoes gelatinisation during baking?** **[K] / [CO2]**
   Ans: Starch
7. **Mention two advantages of using SCP as a protein source?** **[K] / [CO2]**
   Ans: Any two advantages (2 marks)
8. **What is enterotoxin?** **[K] / [CO3]**
   Ans: Definition (1mark) .Any one example (1mark)
9. **How can you prevent food poisoning at home?** **[A] / [CO3]**
   Ans: Any two methods (2 marks)
10. **What conditions favor botulinum toxin production?** **[K] / [CO3]**
    Ans: Any two conditions (2 marks)

11. **What is meant by frozen storage?** **[K] / [CO1]**
Ans: Definition (2 marks)

12. **Name any two food industries where HACCP is applied.** **[K] / [CO3]**
Ans: Sea food industry, Diary industry, Meat producers and processors etc (2 marks)

[2x10 = 20]

PART B

Short Essay Questions
Answer any **4** questions
Each question carries **5** marks

13. **Illustrate the organism responsible for ciguatoxin production and its role in the marine ecosystem.** **[U] / [CO3]**
Ans: Benthic dinoflagellate, Gambierdiscus toxicus (1 mark). Write its role in the marine ecosystem in brief (4 marks).

14. **What are the characteristic features of Mushroom?** **[K] / [CO2]**
Ans: Definition (1 mark). Morphology and reproduction (2 marks), Nutritional characteristics and importance (2 marks).

15. **Define food poisoning and list its common causes.** **[K] / [CO3]**
Ans: Definition (1 mark), causes – bacterial, Fungal, Viral, Toxins (4 marks)

16. **Give an outline of microbial examination of food.** **[K] / [CO1]**
Ans: Sample Collection and Handling, Sample Preparation, Microbial Enumeration, Isolation and Identification of Microorganisms, Detection of Specific Pathogens.(5 marks)

17. **Describe the different types of bacterial spoilage occurring in milk.** **[K] / [CO2]**
Ans: Write any 5 types of spoilage in milk (5 marks)

18. **Explain gas production, proteolysis and ropiness in milk**. **[U] / [CO1]**
Ans: Mechanism and organism involved in gas production, proteolysis and ropiness in milk in brief. (5 marks)

[5x4 = 20]

PART C

ESSAY QUESTION
Answer any **1** question
Each question carries **10** marks

19. **Explain the different types of spoilage in meat.** **[U] / [CO1]**
Ans: Definition of spoilage (1 mark). Different types of spoilage in meat in detail (9 marks)

20. **Explain how nutraceuticals can be used in the management of lifestyle- related disorders.** **[U] / [CO2]**
Ans: Definition (1 mark), Role in Diabetes Mellitus, Role in Cardiovascular Diseases, Role in Obesity Management, Role in Hypertension, Gut Health and Immunity, Antioxidant and Anti-inflammatory Effects. Write the role in any 4 life style diseases (9 marks)